Cheat Sheet: Computational Models (20604)

https://github.com/adielbm/20604

	$\overline{\text{REG}}$	REG	CFL	DEC.	REC.	P	NP	NPC
$L1 \cup L2$	no	✓	✓	✓	✓	✓	✓	no
$L1 \cap L2$	no	✓	no	✓	✓	✓	✓	no
$\overline{L}$	✓	✓	no	✓	no	✓	?	?
$L1 \cdot L2$	no	✓	✓	✓	✓	✓	✓	no
L^*	no	✓	✓	✓	✓	✓	✓	no
$L^{\mathcal{R}}$	✓	✓	✓	✓	✓	✓		
$L1 \setminus L2$	no	✓	no	✓	no	✓	?	
$L \cap R$	no	✓	✓	✓	✓	✓		

- **DFA:** $D = (Q, \Sigma, \delta, q_0, F)$, $\delta : Q \times \Sigma \to Q$.
- **NFA:** $N = (Q, \Sigma, \delta, q_0, F)$, $\delta : Q \times \Sigma_\varepsilon \to \mathcal{P}(Q)$.
- **GNFA:** $(Q, \Sigma, \delta, q_0, q_a), \delta : Q \setminus \{q_a\} \times Q \setminus \{q_0\} \to \text{Reg}(\Sigma)$
- $\forall D_1, D_2, \exists D : |Q| = |Q_1| \cdot |Q_2|$, $L(D) = L(D_1) \Delta L(D_2)$.
- (DFA D) If $L(D) \neq \emptyset$ then $\exists\, s \in L(D)$ s.t. $|s| < |Q|$.

- $\forall$ NFA $\exists$ an equivalent NFA with 1 accept state.
- If $A = L(N_{\text{NFA}}), B = (L(M_{\text{DFA}}))^{\complement}$ then $A \cdot B \in$ REG.

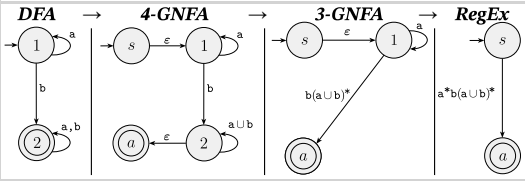
Regular Expressions: Examples

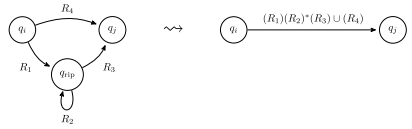
- $\{a^n w b^n : w \in \Sigma^*\} \equiv a(a \cup b)^* b$
- $\{w : \#_w(0) \geq 2 \vee \#_w(1) \leq 1\} \equiv (\Sigma^* 0 \Sigma^* 0 \Sigma^*) \cup (0^*(\varepsilon \cup 1)0^*)$
- $\{w : |w| \bmod n = m\} \equiv (a \cup b)^m ((a \cup b)^n)^*$
- $\{w : \#_b(w) \bmod n = m\} \equiv (a^* b a^*)^m \cdot ((a^* b a^*)^n)^*$
- $\{w : |w| \text{ is odd}\} \equiv (a \cup b)^*((a \cup b)(a \cup b)^*)^*$
- $\{w : \#_a(w) \text{ is odd}\} \equiv b^* a (a b^* a \cup b)^*$
- $\{w : \#_{ab}(w) = \#_{ba}(w)\} \equiv \varepsilon \cup a \cup b \cup a\Sigma^* a \cup b\Sigma^* b$
- $\{a^m b^n \mid m + n \text{ is odd}\} \equiv a(aa)^*(bb)^* \cup (aa)^* b (bb)^*$
- $\{aw : aba \not\subseteq w\} \equiv a(a \cup bb \cup bbb)^*(b \cup \varepsilon)$
- $\{w : bb \not\subseteq w\} \equiv (a \cup ba)^*(\varepsilon \cup b)$
- $\{w : \#_w(a), \#_w(b) \text{ are even}\} \equiv (aa \cup\ bb \cup (ab \cup ba)^2)^*$
- $\{w : |w| \bmod n \neq m\} \equiv \bigcup_{r=0, r \neq m}^{n-1} (\Sigma^n)^* \Sigma^r$

N	δ	a	b	ε
$\to$	1	3	$\{\}$	2
Ⓐ	2	1	$\{\}$	$\{\}$
	3	2	2, 3	$\{\}$

NFA $\to$ DFA

D	δ	a	b
	1	3	$\{\}$
Ⓐ	2	1, 2	$\{\}$
	3	2	2,3
Ⓐ	1,2	1,3,2	$\{\}$
	1,3	2,3	2,3
Ⓐ	2,3	1,2	2,3
Ⓐ	1,2,3	1,2,3	2,3





Pumping lemma for regular languages: $A \in \text{REGULAR} \implies \exists p : \forall s \in A, |s| \geq p, s = xyz$, **(i)** $\forall i \geq 0, xy^i z \in A$, **(ii)** $|y| > 0$ **and (iii)** $|xy| \leq p$.

non-regular but CFL: Examples

- $\{w = w^{\mathcal{R}}\}$; $s = 0^p 1 0^p = xyz$. but $xy^2z = 0^{p+|y|}10^p \notin L$.
- $\{a^n b^n\}$; $s = a^p b^p = xyz$, $xy^2z = a^{p+|y|}b^p \notin L$.
- $\{w : \#_a(w) > \#_b(w)\}$; $s = a^p b^{p+1}$, $|s| = 2p + 1 \geq p$, $xy^2z = a^{p+|y|}b^{p+1} \notin L$.
- $\{w : \#_a(w) = \#_b(w)\}$; $s = a^p b^p = xyz$ but $xy^2z = a^{p+|y|}b^p \notin L$.
- $\{w = w^{\mathcal{R}}\}$; $s = 0^p 1 0^p = xyz$. but $xy^2z = 0^{p+|y|}10^p \notin L$.
- $\{w : \#_w(a) \neq \#_w(b)\}$; (*pf.* by 'complement-closure', $\overline{L} = \{w : \#_w(a) = \#_w(b)\}$)
- $\{a^i b^j c^k : i < j \vee i > k\}$; $s = a^p b^{p+1} c^{2p} = xyz$, but $xy^2z = a^{p+|y|}b^{p+1}c^{2p}$, $p + |y| \geq p + 1$, $p + |y| \leq 2p$.

non-CFL and non-regular: Examples

- $\{w = a^{2^k}\}$; $\quad k = \lfloor \log_2 |w| \rfloor$, $s = a^{2^k} = xyz$. $2^k = |xyz| < |xy^2z| \leq |xyz| + |xy| \leq 2^k + p < 2^{k+1}$.
- $\{a^p : p \text{ is prime}\}$; $\quad s = a^t = xyz$ for prime $t \geq p$. $r := |y| > 0$
- $\{www : w \in \Sigma^*\}$; $s = a^p b a^p b a^p = xyz = a^{|x|+|y|+m} b a^p b a^p b$, $m \geq 0$, but $xy^2z = a^{|x|+2|y|+m} b a^p b a^p b \notin L$.
- $\{a^{2n} b^{3n} a^n\}$; $s = a^{2p} b^{3p} a^p = xyz = a^{|x|+|y|+m+p} b^{3p} a^p$, $m \geq 0$, but $xy^2z = a^{2p+|y|} b^{3p} a^p \notin L$.

(PDA) $M = (Q, \Sigma, \Gamma, \delta, q_0 \in Q, F \subseteq Q)$. $\delta : Q \times \Sigma_\varepsilon \times \Gamma_\varepsilon \longrightarrow \mathcal{P}(Q \times \Gamma_\varepsilon)$. $\quad L \in \textbf{CFL} \Leftrightarrow \exists G_{\text{CFG}} : L = L(G) \Leftrightarrow \exists P_{\text{PDA}} : L = L(P)$

- "$a, b \to c$": **reads** a from the input (or read nothing if $a = \varepsilon$). **pops** b from the stack (or pops nothing if $b = \varepsilon$). **pushes** c onto the stack (or pushes nothing if $c = \varepsilon$)
- If $G \in$ CNF, and $w \in L(G)$, then $|w| \leq 2^{|h|} - 1$, where h is the height of the parse tree for w.
- $\forall L \in$ CFL, $\exists G \in$ CNF $: L = L(G)$.
- (**derivation**) $S \Rightarrow u_1 \Rightarrow u_2 \Rightarrow \cdots \Rightarrow u_n = w$, where each u_i is in $(V \cup \Sigma)^*$. (in this case, G **generates** w (or S **derives** w), $S \overset{*}{\Rightarrow} w$)
- M **accepts** $w \in \Sigma^*$ if there is a seq. $r_0, r_1, \ldots, r_m \in Q$ and $s_0, , s_1, \ldots, s_m \in \Gamma^*$ s.t.: (1.) $r_0 = q_0$ and $s_0 = \varepsilon$; (2.) For $i = 0, 1, \ldots, m-1$, we have $(r_i, b) \in \delta(r_i, w_{i+1}, a)$, where $s_i = at$ and $s_{i+1} = bt$ for some $a, b \in \Gamma_\varepsilon$ and $t \in \Gamma^*$; (3.) $r_m \in F$.
- $R \in \text{REGULAR} \wedge C \in \text{CFL} \implies R \cap C \in$ CFL. (*pf.* construct PDA $P' = P_C \times D_R$.)

(CFG) $G = (V, \Sigma, R, S)$, $A \to w$, $(A \in V, w \in (V \cup \Sigma)^*)$; **(CNF)** $A \to BC$, $A \to a$, $S \to \varepsilon$, $(A, B, C \in V, a \in \Sigma, B, C \neq S)$.

(**CFG** $\rightsquigarrow$ **CNF**) (**1.**) Add a new start variable S_0 and a rule $S_0 \to S$. (**2.**) Remove ε-rules of the form $A \to \varepsilon$ (except for $S_0 \to \varepsilon$). and remove A's occurrences on the RH of a rule (e.g. $R \to uAvAw$ becomes $R \to uAvAw|uAvw|uvAw|uvw$. where $u, v, w \in (V \cup \Sigma)^*$). (**3.**) Remove unit rules $A \to B$ then whenever $B \to u$ appears, add $A \to u$, unless this was a unit rule previously removed. ($u \in (V \cup \Sigma)^*$). (**4.**) Replace each rule $A \to u_1 u_2 \cdots u_k$ where $k \geq 3$ and $u_i \in (V \cup \Sigma)$, with the rules $A \to u_1 A_1$, $A_1 \to u_2 A_2$, ..., $A_{k-2} \to u_{k-1} u_k$, where A_i are new variables. Replace terminals u_i with $U_i \to u_i$.

CFL but non-regular: Examples

- $\{w : w = w^{\mathcal{R}}\}; S \to aSa \mid bSb \mid a \mid b \mid \varepsilon$
- $\{w : w \neq w^{\mathcal{R}}\}; S \to aSa \mid bSb \mid aXb \mid bXa; X \to aX|bX|\varepsilon$
- $\{ww^{\mathcal{R}}\} = \{w : w = w^{\mathcal{R}} \wedge |w| \text{ is even}\}; S \to aSa \mid bSb \mid \varepsilon$
- $\overline{\{ww^{\mathcal{R}}\}}; S \to aSa \mid bSb \mid aXb \mid bXa \mid a \mid b;$ $X \to aXa \mid bXb \mid bXa \mid aXb \mid a \mid b \mid \varepsilon$
- $\{wa^n w^{\mathcal{R}}\}; S \to aSa \mid bSb \mid M; M \to aM \mid \varepsilon$
- $\{w\#x : w^{\mathcal{R}} \subseteq x\}; S \to AX;\ A \to 0A0 \mid 1A1 \mid \#X;$ $X \to 0X \mid 1X \mid \varepsilon$
- $\{w : \#_w(a) > \#_w(b)\}; S \to IaI; I \to II \mid aIb \mid bIa \mid a \mid \varepsilon$
- $\{w : \#_w(a) \geq \#_w(b)\}; S \to SS \mid aSb \mid bSa \mid a \mid \varepsilon$
- $\{w : \#_w(a) = \#_w(b)\}; S \to SS \mid aSb \mid bSa \mid \varepsilon$
- $\{w : \#_w(a) = 2 \cdot \#_w(b)\};$ $S \to SS|S_1bS_1|bSaa|aaSb|\varepsilon; S_1 \to aS|SS_1$
- $\{w : \#_w(a) \neq \#_w(b)\} = \{\#_w(a) > \#_w(b)\} \cup \{\#_w(a) < \#_w(b)\}$
- $\overline{\{a^n b^n\}}; S \to XbXaX \mid A \mid B;\ A \to aAb \mid Ab \mid b;$ $B \to aBb \mid aB \mid a;\ X \to aX \mid bX \mid \varepsilon.$
- $\{a^n b^m \mid n \neq m\}; S \to aSb|A|B; A \to aA|a; B \to bB|b$
- $\{x \mid x \neq ww\}; S \to A \mid B \mid AB \mid BA; A \to CAC \mid 0;$ $B \to CBC \mid 1; C \to 0 \mid 1$
- $\{a^n b^m \mid m \leq n \leq 3m\}; S \to aSb \mid aaSb \mid aaaSb \mid \varepsilon;$
- $\{a^n b^n\}; S \to aSb \mid \varepsilon$
- $\{a^n b^m \mid n > m\}; S \to aSb \mid aS \mid a$
- $\{a^n b^m \mid n \geq m \geq 0\}; S \to aSb \mid aS \mid a \mid \varepsilon$
- $\{a^i b^j c^k \mid i + j = k\}; S \to aSc \mid X; X \to bXc \mid \varepsilon$
- $\{a^i b^j c^k \mid i \leq j \vee j \leq k\}; S \to S_1C \mid AS_2; A \to Aa \mid \varepsilon;$ $S_1 \to aS_1b \mid S_1b \mid \varepsilon; S_2 \to bS_2c \mid S_2c \mid \varepsilon; C \to Cc \mid \varepsilon$
- $\{a^i b^j c^k \mid i = j \vee j = k\};$ $S \to AX_1|X_2C; X_1 \to bX_1c|\varepsilon; X_2 \to aX_2b|\varepsilon; A \to aA|\varepsilon; C$
- $\{xy : |x| = |y|, x \neq y\}; S \to AB \mid BA;$ $A \to a \mid aAa \mid aAb \mid bAa \mid bAb;$ $B \to b \mid aBa \mid aBb \mid bBa \mid bBb;$
- $\{a^i b^j : i, j \geq 1,\ i \neq j,\ i < 2j\};$ $S \to aSb|X|aaYb; Y \to aaYb|ab; X \to bX|abb$

CFL and regular: Examples

- $\{w : \#_w(a) \geq 3\}; S \to XaXaXaX; X \to aX \mid bX \mid \varepsilon$
- $\{w : |w| \text{ is odd}\}; S \to aaS \mid abS \mid bbS \mid baS \mid a \mid b$
- $\{w : |w| \text{ is even}\}; S \to aaS \mid abS \mid bbS \mid baS \mid \varepsilon$
- $\emptyset; S \to S$

Pumping lemma for context-free languages: $L \in \text{CFL} \implies \exists p : \forall s \in L, |s| \geq p, s = uvxyz$, **(i)** $\forall i \geq 0, uv^i xy^i z \in L$, **(ii)** $|vxy| \leq p$, **and (iii)** $|vy| > 0$.

- $\{w = a^n b^n c^n\}$; $s = a^p b^p b^p = uvxyz$. vxy can't contain all of a, b, c thus uv^2xy^2z must pump one of them less than the others.
- $\{ww : w \in \{a, b\}^*\}$;
- (**more example of not CFL**)
- $\{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$, $\{a^n b^n c^n \mid n \in \mathbb{N}\}$, $\{ww \mid w \in \{a, b\}^*\}$, $\{\mathtt{a}^{n^2} \mid n \geq 0\}$, $\{a^p \mid p \text{ is prime}\}$, $L = \{ww^{\mathcal{R}}w : w \in \{a, b\}^*\}$
- $\{w \mid \#_w(a) = \#_w(b) = \#_w(c)\}$: (*pf.* since Regular $\cap$ CFL $\in$ CFL, but $\{a^* b^* c^*\} \cap L = \{a^n b^n c^n\} \notin$ CFL)

Examples

- $A \leq_{\text{m}} B$, $B \in \text{REGULAR}$, $A \notin \text{REGULAR}$: $A = \{0^n 1^n\}$, $B = \{1\}$, $f : A \to B$, $f(w) = 1$ if $w \in A$, 0 if $w \notin A$.
- $L \in$ CFL, $\overline{L} \notin$ CFL: $L = \{x \mid x \neq ww\}$, $\overline{L} = \{ww\}$.
- $L_1, L_2 \in$ CFL, $L_1 \cap L_2 \notin$ CFL: $L_1 = \{a^n b^n c^m\}$, $L_2 = \{a^m b^n c^n\}$, $L_1 \cap L_2 = \{a^n b^n c^n\}$.
- $L_1, L_2 \notin$ CFL, $L_1 \cap L_2 \in$ CFL: $L_1 = \{a^n b^n c^n\}$, $L_2 = \{c^n b^n a^n\}$, $L_1 \cap L_2 = \{\varepsilon\}$
- $L_1 \in$ CFL, $L_2, L_1 \cap L_2 \notin$ CFL: $L_1 = \Sigma^*$, $L_2 = \{a^{i^2}\}$.
- $L_1 \in \text{REGULAR}$, $L_2 \notin$ CFL, but $L_1 \cap L_2 \in$ CFL : $L_1 = \{\varepsilon\}$, $L_2 = \{a^n b^n c^n \mid n \geq 0\}$.
- $L_1 \in$ CFL, L_2 is infinite, $L_1 \setminus L_2 \notin \text{REGULAR}$: $L_1 = \Sigma^*$, $L_2 = \{a^n b^n\}$, $L_1 \setminus L_2 = \{a^m b^n \mid m \neq n\}$.
- $L_1, L_2 \in \text{REGULAR}$, $L_1 \not\subset L_2$, $L_2 \not\subset L_1$, but, $(L_1 \cup L_2)^* = L_1^* \cup L_2^*$: $L_1 = \{a, b, ab\}$, $L_2 = \{a, b, ba\}$.
- $L_1, L_1 \cup L_2 \in \text{REGULAR}$, $L_2, L_1 \cap L_2 \notin \text{REGULAR}$, $L_1 = L(\mathtt{a}^*\mathtt{b}^*)$, $L_2 = \{\mathtt{a}^n \mathtt{b}^n \mid n \geq 0\}$.
- $L_1, L_2, \cdots \in \text{REGULAR}$, $\bigcup_{i=1}^{\infty} L_i \notin \text{REGULAR}$: $L_i = \{\mathtt{a}^i \mathtt{b}^i\}$, $\bigcup_{i=1}^{\infty} L_i = \{\mathtt{a}^n \mathtt{b}^n \mid n \geq 0\}$.
- $L_1 \cdot L_2 \in \text{REGULAR}$, $L_1 \notin$ Reg. : $L_1 = \{a^n b^n\}$, $L_2 = \Sigma^*$
- $L_2 \in$ CFL, and $L_1 \subseteq L_2$, but $L_1 \notin$ CFL : $\quad \Sigma = \{a, b, c\}$, $L_1 = \{a^n b^n c^n \mid n \geq 0\}$, $L_2 = \Sigma^*$.
- $L_1, L_2 \in$ TD, and $L_1 \subseteq L \subseteq L_2$, but $L \notin$ TD : $\quad L_1 = \emptyset$, $L_2 = \Sigma^*$, L is some undecidable language over Σ.
- $L^* \in \text{REGULAR}$, but $L \notin \text{REGULAR}$: $L = \{a^p \mid p \text{ is prime}\}$, $L^* = \Sigma^* \setminus \{a\}$.
- $A \not\leq_m \overline{A} : A = A_{\text{TM}} \in$ TR, $\overline{A} = \overline{A_{\text{TM}}} \notin$ TR
- $A \notin$ DEC., $A \leq_m \overline{A} : f(0x) = 1x, f(1y) = 0y$, $A = \{w \mid \exists x \in A_{\text{TM}} : w = 0x \vee \exists y \in \overline{A_{\text{TM}}} : w = 1y\}$
- $L \in$ CFL, $L \cap L^{\mathcal{R}} \notin$ CFL : $L = \{a^n b^n a^m\}$.
- $A \leq_m B, B \not\leq_m A : A = \{a\}, B = HALT_{\text{TM}}, f(w) = \langle M \rangle$, $M =$"On x, if $w \in A$, Ⓐ; O/W, loop"

$L \in$ **T**uring-**D**ecidable $\iff$ ($L \in$ **T**uring-**R**ecognizable **and** $\overline{L} \in$ **T**uring-**R**ecognizable) $\iff \exists M_{\mathsf{TM}}$ decides L. REGULAR $\subset$ CFL $\subset$ TD $\subset$ TR.

- **(TM)** $M = (Q, \underset{\text{input}}{\Sigma} \subseteq \Gamma, \underset{\text{tape}}{\Gamma}, \delta, q_0, q_{\text{A}}, q_{\boxed{\text{R}}})$, where $\sqcup \in \Gamma$, $\sqcup \notin \Sigma$, $q_{\boxed{\text{R}}} \neq q_{\text{A}}$, $\delta : Q \times \Gamma \longrightarrow Q \times \Gamma \times \{\text{L}, \text{R}\}$
- **(Turing-Recognizable (TR))** A if $w \in L$, $\boxed{\text{R}}$/loops if $w \notin L$; A is **co-recognizable** if $\overline{A}$ is recognizable.
- **(Turing-Decidable (TD))** A if $w \in L$, $\boxed{\text{R}}$ if $w \notin L$.
- $L \in \text{TR} \iff L \leq_{\text{m}} A_{\mathsf{TM}}$.
- $(A \in \text{TR} \wedge |A| = \infty) \Rightarrow \exists B \in \text{TD} : (B \subseteq L \wedge |B| = \infty)$
- **(Rice)** If $P = \{\langle M \rangle : L(M) \text{ has property } \mathcal{P}\}$ s.t. (**1**) $\forall M_1, M_2 : L(M_1) = L(M_2) \Rightarrow (\langle M_1 \rangle \in P \Leftrightarrow \langle M_2 \rangle \in P)$. (**2**) P is nontrivial. Then $P \notin$ TD. (*e.g.* $INFINITE_{\mathsf{TM}}$, ALL_{TM}, E_{TM}, $\{\langle M_{\mathsf{TM}} \rangle : 1 \in L(M)\}$)
- {all TMs} is count.; Σ^* is count. (finite Σ); {all lang.} is uncount.; {all infinite bin. seq.} is uncount.
- If $A \leq_{\text{m}} B$ and $B \in$ TD, then $A \in$ TD.
- If $A \leq_{\text{m}} B$ and $A \notin$ TD, then $B \notin$ TD.
- If $A \leq_{\text{m}} B$ and $B \in$ TR, then $A \in$ TR.
- If $A \leq_{\text{m}} B$ and $A \notin$ TR, then $B \notin$ TR.
- (transitivity) If $A \leq_{\text{m}} B$ and $B \leq_{\text{m}} C$, then $A \leq_{\text{m}} C$.
- $A \leq_{\text{m}} B \iff \overline{A} \leq_{\text{m}} \overline{B}$ (esp. $A \leq_{\text{m}} \overline{A} \iff \overline{A} \leq_{\text{m}} A$)
- If $A \leq_{\text{m}} \overline{A}$ and $A \in$ TR, then $A \in$ TD
- **(not TR)** $\overline{A_{\mathsf{TM}}}$, $\overline{EQ_{\mathsf{TM}}}$, EQ_{CFG}, $\overline{HALT_{\mathsf{TM}}}$, REG_{TM}, E_{TM}, EQ_{TM}, ALL_{CFG}, EQ_{CFG}

- **(TR, but not TD)** A_{TM}, $HALT_{\mathsf{TM}}$, $\overline{EQ_{\mathsf{CFG}}}$, $\overline{E_{\mathsf{TM}}}$, $\{\langle M, k \rangle \mid \exists x\ (M(x) \text{ halts in } \geq k \text{ steps})\}$
- **(TD)** A_{DFA}, A_{NFA}, A_{REX}, E_{DFA}, EQ_{DFA}, A_{CFG}, E_{CFG}, A_{LBA}

Deciders (TM that halts on all inputs): Examples

- $INFINITE_{\mathsf{DFA}}$: "On $\langle D \rangle$: $n := |Q_D|$; const. D_1 s.t. $L(D_1) = \Sigma^{\geq n}$; const. D_2 s.t. $L(D_2) = L(D) \cap L(D_1)$; if $\langle D_2 \rangle \notin E_{\mathsf{DFA}}$, A; O/W, $\boxed{\text{R}}$"
- EQ_{DFA}: "On $\langle D_1, D_2 \rangle$: const. D s.t. $L(D) = (L(D_1) \cap \overline{L(D_2)}) \cup (\overline{L(D_1)} \cap L(D_2))$; if $\langle D \rangle \in E_{\mathsf{DFA}}$, A; O/W, $\boxed{\text{R}}$"
- ALL_{DFA}: "On $\langle D \rangle$: const. $D^{\complement}$ s.t. $L(D^{\complement}) = L(D)^{\complement}$ (swap accept and non-accept); if $D^{\complement} \in E_{\mathsf{DFA}}$, A; O/W $\boxed{\text{R}}$"
- $\{\langle D \rangle \mid \nexists w \in L(D) : \#_1(w) \text{ is odd}\}$: "On $\langle D \rangle$: const. D_1 s.t. $L(D_1) = \{w \mid \#_1(w) \text{ is odd}\}$; const. D_2 s.t. $L(D_2) = L(D) \cap L(D_1)$; if $\langle D_2 \rangle \in E_{\mathsf{DFA}}$ A; O/W $\boxed{\text{R}}$"
- $\{\langle r, s \rangle \mid r, s \in \text{Reg}(\Sigma), L(r) \subseteq L(s)\}$: "On $\langle r, s \rangle$: const. D s.t. $L(D) = L(r) \cap \overline{L(s)}$; if $\langle D \rangle \in E_{\mathsf{DFA}}$, A; O/W, $\boxed{\text{R}}$"
- $\{\langle D, r \rangle \mid L(D) = L(r)\}$: "On $\langle D, r \rangle$: convert r to DFA D_r; if $\langle D, D_r \rangle \in EQ_{\mathsf{DFA}}$, A; O/W, $\boxed{\text{R}}$"
- $\{\langle D_{\mathsf{DFA}} \rangle \mid L(D) = (L(D))^{\mathcal{R}}\}$: "On $\langle D \rangle$: const. $D^{\mathcal{R}}$ s.t. $L(D^{\mathcal{R}}) = (L(D))^{\mathcal{R}}$; if $\langle D, D^{\mathcal{R}} \rangle \in EQ_{\mathsf{DFA}}$), A; O/W, $\boxed{\text{R}}$"
- $\{\langle r \rangle \mid \exists x, y \in \Sigma^* : w = x111y \in L(r)\}$: "On $\langle r \rangle$: const. D s.t. $L(D) \equiv \Sigma^*111\Sigma^*$; const. D_1 s.t. $L(D_1) = L(r) \cap L(D)$; if $L(D_1) \notin E_{\mathsf{DFA}}$, A; O/W $\boxed{\text{R}}$"
- $\{\langle G, x \rangle \mid \exists y \in L(G) : x \subseteq y\}$: "On $\langle G, x \rangle$: set P s.t. $L(P) = L(G)$; set $P_x(y) := P(xy)$; If $\langle P_x \rangle \notin E_{\mathsf{PDA}}$, A; O/W $\boxed{\text{R}}$"
- $\{\langle G, k \rangle : |L(G)| = k \in \mathbb{N} \cup \{\infty\}\}$: "On $\langle G, k \rangle$: run ; if $\langle G \rangle \in INFINITE_{\mathsf{CFG}}$: (if $k = \infty$, A; O/W, $\boxed{\text{R}}$). if $\langle G \rangle \notin INFINITE_{\mathsf{CFG}}$: (if $k = \infty$, $\boxed{\text{R}}$; O/W, m counts each $w \in \Sigma^{\leq p}$ s.t. $w \in L(G)$, where p is the pump. len.; if $m = k$, A, O/W, $\boxed{\text{R}}$)
- $A\varepsilon_{\mathsf{CFG}}$: "On $\langle G \rangle$: If $\langle G, \varepsilon \rangle \in A_{\mathsf{CFG}}$, A; O/W, $\boxed{\text{R}}$"
- $INFINITE_{\mathsf{PDA}}$: "On $\langle P \rangle$: conv. P to G; $p :=$ p.l. of G; set $G' \equiv L(G') = L(G) \cap \Sigma^{>p}$; If $\langle G' \rangle \notin E_{\mathsf{CFG}}$, A; O/W $\boxed{\text{R}}$"
- $\{\langle G \rangle : 1^* \cap L(G) \neq \emptyset\}$; "On $\langle G \rangle$: const. G' s.t. $L(G') = 1^* \cap L(G)$. (since REGULAR $\cap$ CFL $\subseteq$ CFL); If $\langle G' \rangle \notin E_{\mathsf{CFG}}$, A; O/W, $\boxed{\text{R}}$"
- $\{\langle M, k \rangle \mid \exists x\ (M(x) \text{ runs for } \geq k \text{ steps})\}$: "On $\langle M, k \rangle$: ($\forall w \in \Sigma^{\leq k+1}$: if $M(w)$ not halt within k steps, A); $\boxed{\text{R}}$"
- $\{\langle M, k \rangle \mid \exists x\ (M(x) \text{ halts in } \leq k \text{ steps})\}$: "On $\langle M, k \rangle$: ($\forall w \in \Sigma^{\leq k+1}$: run $M(w)$ for $\leq k$ steps, if halts, A); $\boxed{\text{R}}$"

Recognizers: Examples

- $\overline{EQ_{\mathsf{CFG}}}$: "On $\langle G_1, G_2 \rangle$: (for each $w \in \Sigma^*$ (lexico.): If $\langle G_1, w \rangle \in A_{\mathsf{CFG}}$ and $\langle G_2, w \rangle \notin A_{\mathsf{CFG}}$ (vice versa), A);"
- $\overline{E_{\mathsf{TM}}}$: "On $\langle M \rangle$: $\Sigma^* = \{s_1, s_2, \ldots\}$; $\forall i \in \mathbb{N}$: $\forall j \leq i$: Run $M(s_j)$ for i steps, if accepts, A;"

Mapping Reduction (from A to B): $A \leq_{\text{m}} B$ if $\exists f : \Sigma^* \to \Sigma^* : \forall w \in \Sigma^*$, $w \in A \iff f(w) \in B$ and f is computable.

- $A_{\mathsf{TM}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle \mid L(M) = (L(M))^{\mathcal{R}}\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x, if $x \notin \{01, 10\}$, $\boxed{\text{R}}$; if $x = 01$, return $M(x)$; if $x = 10$, A;"
- $A_{\mathsf{TM}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle \mid \varepsilon \in L(M)\}$; $f(\langle M, w \rangle) = \langle M' \rangle$ where $M' =$"On x, if $x \neq \varepsilon$, A; O/W return $M(w)$"
- $A_{\mathsf{TM}} \leq_{\text{m}} L = \{\langle \underset{\text{TM}}{M}, \underset{\text{DFA}}{D} \rangle \mid L(M) = L(D)\}$; $f(\langle M, w \rangle) = \langle M', D \rangle$, where $M' =$"On x: if $x = w$ return $M(x)$; O/W, $\boxed{\text{R}}$;" D is DFA s.t. $L(D) = \{w\}$.
- $A \leq_{\text{m}} HALT_{\mathsf{TM}}$; $f(w) = \langle M, \varepsilon \rangle$, where $M =$"On x: if $w \in A$, halt; if $w \notin A$, loop;"
- $A_{\mathsf{TM}} \leq_{\text{m}} \{\langle M \rangle \mid L(M) \text{ is CFL}\}$; $f(\langle M, w \rangle) = \langle N \rangle$, where $N =$"On x: if $x = a^n b^n c^n$, A; O/W, return $M(w)$;"
- $A \leq_{\text{m}} B = \{0w : w \in A\} \cup \{1w : w \notin A\}$; $f(w) = 0w$.
- $A_{\mathsf{TM}} \leq_{\text{m}} HALT_{\mathsf{TM}}$; $f(\langle M, w \rangle) = \langle M', w \rangle$, where $M' =$ "On x: if $M(x)$ accepts, A. If rejects, loop"
- $HALT_{\mathsf{TM}} \leq_{\text{m}} A_{\mathsf{TM}}$; $f(\langle M, w \rangle) = \langle M', \langle M, w \rangle \rangle$, where $M' =$"On $\langle X, x \rangle$: if $X(x)$ halts, A;"

- $E_{\mathsf{TM}} \leq_{\text{m}} USELESS_{\mathsf{TM}}$; $f(\langle M \rangle) = \langle M, q_{\text{A}} \rangle$
- $E_{\mathsf{TM}} \leq_{\text{m}} EQ_{\mathsf{TM}}$; $f(\langle M \rangle) = \langle M, M' \rangle$, $M' =$"On x: $\boxed{\text{R}}$"
- $A_{\mathsf{TM}} \leq_{\text{m}} REGULAR_{\mathsf{TM}}$; $f(\langle M, w \rangle) = \langle M' \rangle$, $M' =$"On $x \in \{0, 1\}^*$: if $x = 0^n 1^n$, A; O/W, return $M(w)$;"
- $A_{\mathsf{TM}} \leq_{\text{m}} EQ_{\mathsf{TM}}$; $f(\langle M, w \rangle) = \langle M_1, M_2 \rangle$, where $M_1 =$ "A all"; $M_2 =$"On x: return $M(w)$;"
- $A_{\mathsf{TM}} \leq_{\text{m}} \overline{EQ_{\mathsf{TM}}}$; $f(\langle M, w \rangle) = \langle M_1, M_2 \rangle$, where $M_1 =$ "$\boxed{\text{R}}$ all"; $M_2 =$"On x: return $M(w)$;"
- $A_{\mathsf{TM}} \leq_{\text{m}} \{\langle M \rangle : M \text{ halts on } \langle M \rangle\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: if $M(w)$ accepts, A; if rejects, loop;"
- $ALL_{\mathsf{CFG}} \leq_{\text{m}} EQ_{\mathsf{CFG}}$; $f(\langle G \rangle) = \langle G, H \rangle$, s.t. $L(H) = \Sigma^*$.
- $A_{\mathsf{TM}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle : |L(M)| = 1\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: if $x = x_0$, return $M(w)$; O/W, $\boxed{\text{R}}$;" (where $x_0 \in \Sigma^*$ is fixed).
- $\overline{A_{\mathsf{TM}}} \leq_{\text{m}} E_{\mathsf{TM}}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: if $x \neq w$, $\boxed{\text{R}}$; O/W, return $M(w)$;"

- $\overline{HALT_{\mathsf{TM}}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle : |L(M)| \leq 3\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: A if $M(w)$ halts"
- $HALT_{\mathsf{TM}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle : |L(M)| \geq 3\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: A if $M(w)$ halts"
- $\overline{HALT_{\mathsf{TM}}} \leq_{\text{m}} \{\langle M \rangle : M \text{ A even num.}\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, $M' =$"On x: $\boxed{\text{R}}$ if $M(w)$ halts within $|x|$. O/W, A"
- $\overline{HALT_{\mathsf{TM}}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle : L(M) \text{ is finite}\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: A if $M(w)$ halts"
- $\overline{HALT_{\mathsf{TM}}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle : L(M) \text{ is infinite}\}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: $\boxed{\text{R}}$ if $M(w)$ halts within $|x|$ steps. O/W, A"
- $HALT_{\mathsf{TM}} \leq_{\text{m}} \{\langle M_1, M_2 \rangle : \varepsilon \in L(M_1) \cup L(M_2)\}$; $f(\langle M, w \rangle) = \langle M', M' \rangle$, $M' =$"On x: A if $M(w)$ halts"
- $HALT_{\mathsf{TM}} \leq_{\text{m}} \overline{E_{\mathsf{TM}}}$; $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: if $x \neq w$ $\boxed{\text{R}}$; else, A if $M(w)$ halts"
- $HALT_{\mathsf{TM}} \leq_{\text{m}} \{\langle M_{\mathsf{TM}} \rangle \mid \exists x : M(x) \text{ halts in } > |\langle M \rangle| \text{ steps}\}$ $f(\langle M, w \rangle) = \langle M' \rangle$, where $M' =$"On x: if $M(w)$ halts, make $|\langle M \rangle| + 1$ steps and then halt; O/W, loop"

$\mathbf{P} = \bigcup_{k \in \mathbb{N}} \text{TIME}(n^k) \subseteq \mathbf{NP} = \bigcup_{k \in \mathbb{N}} \text{NTIME}(n^k) = \{L \mid L \text{ is decidable by a PT verifier}\} \supseteq$ **NP-complete** $= \{B \mid B \in \text{NP}, \forall A \in \text{NP}, A \leq_{\text{P}} B\}$.

- If $A \leq_{\text{P}} B$ and $B \in \text{P}$, then $A \in \text{P}$.
- $A \equiv_P B$ if $A \leq_{\text{P}} B$ and $B \leq_{\text{P}} A$. $\equiv_P$ is an equiv. relation on NP. $\text{P} \setminus \{\emptyset, \Sigma^*\}$ is an equiv. class of $\equiv_P$.
- ALL_{DFA}, $CONNECTED$, $\underset{\text{3-clique}}{TRIANGLE}$, $L(G_{\mathsf{CFG}})$, $\underset{s \to t}{\overset{\text{directed}}{PATH}} \in \text{P}$

- $CNF_2 \in \text{P}$: (**algo.** $\forall x \in \phi$: (**1**) If x occurs 1-2 times in same clause $\to$ remove cl.; (**2**) If x is twice in 2 cl. $\to$ remove both cl.; (**3**) Similar to (2) for $\overline{x}$; (**4**) Replace any $(x \vee y)$, $(\neg x \vee z)$ with $(y \vee z)$; (y, z may be ε); (**5**) If $(x) \wedge (\neg x)$ found, $\boxed{\text{R}}$. (**6**) If $\phi = \varepsilon$, A;)

- $CLIQUE$, $SUBSET\text{-}SUM$, SAT, $3SAT$, $\overset{VERTEX}{COVER}$, $HAMPATH$, $UHAMATH$, $3COLOR \in$ NP-complete. $\emptyset, \Sigma^* \notin$ NP-complete.
- If $B \in$ NP-complete and $B \in \text{P}$, then $\text{P} = \text{NP}$.
- If $B \in$ NPC and $C \in$ NP s.t. $B \leq_{\text{P}} C$, then $C \in$ NPC.
- If $\text{P} = \text{NP}$, then $\forall A \in \text{P} \setminus \{\emptyset, \Sigma^*\}$, $A \in$ NP-complete.

Polytime Reduction (from A to B): $A \leq_{\text{P}} B$ if $\exists f : \Sigma^* \to \Sigma^* : \forall w \in \Sigma^*$, $w \in A \iff f(w) \in B$ and f is polytime computable.

- $SAT \leq_{\text{P}} DOUBLE\text{-}SAT$; $f(\phi) = \phi \wedge (x \vee \neg x)$
- $3SAT \leq_{\text{P}} 4SAT$; $f(\phi) = \phi'$, where ϕ' is obtained from the 3cnf ϕ by adding a new var. x to each clause, and adding a new clause $(\neg x \vee \neg x \vee \neg x \vee \neg x)$.
- $3SAT \leq_{\text{P}} CNF_3$; $f(\langle \phi \rangle) = \phi'$. If $\#_\phi(x) = k > 3$, replace x with $x_1, \ldots x_k$, and add $(\overline{x_1} \vee x_2) \wedge \cdots \wedge (\overline{x_k} \vee x_1)$.
- $3SAT \leq_{\text{P}} CLIQUE$; $f(\phi) = \langle G, k \rangle$. where ϕ is 3cnf with k clauses. Nodes represent literals. Edges connect all pairs except those 'from the same clause' or 'contradictory literals'.
- $SUBSET\text{-}SUM \leq_{\text{P}} SET\text{-}PARTITION$; $f(\langle x_1, \ldots, x_m, t \rangle) = \langle x_1, \ldots, x_m, S - 2t \rangle$, where S sum of $x_1, \ldots, x_m$, and t is the target subset-sum.
- $3SAT \leq_{\text{P}} \overset{\text{almost}}{3SAT}$; $f(\phi) = \phi' = \phi \wedge (x \vee x \vee x) \wedge (\overline{x} \vee \overline{x} \vee \overline{x})$
- $3COLOR \leq_{\text{P}} \overset{\text{almost}}{3COLOR}$; $f(\langle G \rangle) = \langle G' = G \sqcup G \rangle$
- $\overset{VERTEX}{COVER}_k \leq_{\text{P}} WVC$; $f(\langle G, k \rangle) = (G, w, k), \forall v \in V, w(v) = 1$
- (dir.) $HAM\text{-}PATH \leq_{\text{P}} 2HAM\text{-}PATH$; $f(\langle G, s, t \rangle) = \langle G', s', t' \rangle$, $V' = V \cup \{s', t', a, b, c, d\}$, $E' = E \cup \{(s', a), (a, b), (b, s)\} \cup \{(s', b), (b, a), (a, s)\} \cup \{(t, c), (c, d), (d, t')\} \cup \{(t, d), (d, c), (c, t')\}$.
- (undir.) $CLIQUE_k \leq_{\text{P}} \underset{|V|/2\text{-clique}}{HALF\text{-}CLIQUE}$; $f(\langle G = (V, E), k \rangle) = \langle G' = (V', E') \rangle$, if $k = \frac{|V|}{2}$, $E = E'$, $V' = V$. if $k > \frac{|V|}{2}$, $V' = V \cup \{j = 2k - |V| \text{ new nodes}\}$. if $k < \frac{|V|}{2}$, $V' = V \cup \{j = |V| - 2k \text{ new nodes}\}$ and $E' = E \cup \{\text{edges for new nodes}\}$
- $\underset{s \rightsquigarrow t}{HAM\text{-}PATH} \leq_{\text{P}} HAM\text{-}CYCLE$; $f(\langle G, s, t \rangle) = \langle G', s, t \rangle$, $V' = V \cup \{x\}$, $E' = E \cup \{(t, x), (x, s)\}$
- $HAM\text{-}CYCLE \leq_{\text{P}} UHAMCYCLE$; $f(\langle G \rangle) = \langle G' \rangle$. For each $u, v \in V$: u is replaced by $u_{\text{in}}, u_{\text{mid}}, u_{\text{out}}$; (v, u) replaced by $\{v_{\text{out}}, u_{\text{in}}\}, \{u_{\text{in}}, u_{\text{mid}}\}$; and (u, v) by $\{u_{\text{out}}, v_{\text{in}}\}, \{u_{\text{mid}}, u_{\text{out}}\}$.
- $UHAMPATH \leq_{\text{P}} PATH_{\geq k}$; $f(\langle G, a, b \rangle) = \langle G, a, b, k = |V| - 1 \rangle$
- $\overset{VERTEX}{COVER} \leq_{\text{P}} CLIQUE$; $f(\langle G, k \rangle) = \langle G^{\complement} = (V, E^{\complement}), |V| - k \rangle$
- $CLIQUE_k \leq_{\text{P}} \{\langle G, t \rangle : G \text{ has } 2t\text{-clique}\}$; $f(\langle G, k \rangle) = \langle G', t = \lceil k/2 \rceil \rangle$, $G' = G$ if k is even; $G' = G \cup \{v\}$ (v connected to all G nodes) if k is odd.
- $CLIQUE_k \leq_{\text{P}} \overset{\text{almost}}{CLIQUE}_k$; $f(\langle G, k \rangle) = \langle G', k + 2 \rangle$, $G' = G \cup \{v_{n+1}, v_{n+2}\}$; v_{n+1}, v_{n+2} are con. to all V
- $\overset{VERTEX}{COVER}_k \leq_{\text{P}} DOMINATING\text{-}SET_k$; $f(\langle G, k \rangle) = \langle G', k \rangle$, where $V' = \{\text{non-isolated nodes in } V\} \cup \{v_e : e \in E\}$, $E' = E \cup \{(v_e, u), (v_e, w) : e = (u, w) \in E\}$.
- $CLIQUE \leq_{\text{P}} INDEP\text{-}SET$; $f(\langle G, k \rangle) = \langle G^{\complement}, k \rangle$
- $\overset{VERTEX}{COVER} \leq_{\text{P}} \underset{\langle \mathcal{U}, \mathcal{S}, k \rangle}{\overset{SET}{COVER}} = \{\exists \mathcal{C} \subseteq \mathcal{S}, |\mathcal{C}| \leq k, \bigcup_{A \in \mathcal{C}} A = \mathcal{U}\}$; $f(\langle G, k \rangle) = \langle \mathcal{U} = E, \mathcal{S} = \{S_1, \ldots, S_n\}, k \rangle$, where $n = |V|$, $S_u = \{\text{edges incident to } u \in V\}$.
- $INDEP\text{-}SET \leq_{\text{P}} \overset{VERTEX}{COVER}$; $f(\langle G, k \rangle) = \langle G, |V| - k \rangle$
- $\overset{VERTEX}{COVER} \leq_{\text{P}} INDEP\text{-}SET$; $f(\langle G, k \rangle) = \langle G, |V| - k \rangle$
- $HAM\text{-}CYCLE \leq_{\text{P}} \{\langle G, w, k \rangle : \exists \text{ hamcycle of weight} \leq k\}$; $f(\langle G \rangle) = \langle G', w, 0 \rangle$, where $G' = (V, E')$, $E' = \{(u, v) \in E : u \neq v\}$, $w(u, v) = 1$ if $(u, v) \in E$, $w(u, v) = 0$ if $(u, v) \notin E$.
- $3COLOR \leq_{\text{P}} SCHEDULE$; $f(\langle G \rangle) = \langle F = V, S = E, h = 3 \rangle$

I. Decidability & Recognizability

- **1. The Enumerator Trap**: **Q:** Does an enumerator exist that prints machines for every recognizable language over **{0,1}**, but those machines *never* halt on rejected words? **A:** Yes. We don't try to solve the Halting Problem to find them. Instead, we enumerate all TMs, and blindly replace any transition to q_{reject} (or undefined crashes) with an infinite loop. **Key Insight:** You don't have to analyze machine behavior to build an enumerator. You can just algorithmically modify source code to force the behavior you want while preserving the language.
- **2. The Mixed Language Recognizer**: **Q:** If $L \in \mathrm{RE}$ and $\overline{L} \notin \mathrm{RE}$, is $L' = \{0w \mid w \in L\} \cup \{1w \mid w \notin L\}$ recognizable? **A:** No. If L' is recognizable (via machine M_1), we can build a recognizer for $\overline{L}$ by simply feeding $1w$ into M_1. Since $\overline{L}$ is strictly not recognizable, this is a contradiction. **Key Insight:** Prove undecidability implicitly by showing that a decider/recognizer for a new language could be used as a trivial subroutine to solve a known impossible language.

II. Mapping Reductions

- **3. The Halting-to-Acceptance Blueprint**: **Q:** Show a mapping reduction $HALT_{\mathsf{TM}} \leq_m A_{\mathsf{TM}}$. **A:** The function $f(\langle M, w\rangle)$ outputs M'. M' runs M on w. If M accepts, M' accepts. If M explicitly rejects, M' accepts. **Key Insight:** In a mapping reduction, f just writes "dumb" source code. Never assume a decider exists inside the generated dummy machine. Halting simply means catching both the accept and reject states.
- **4. The Meta Halting Set**: **Q:** Prove $DEC\text{-}HALT_{\mathsf{TM}}$ is not Turing-Recognizable. **A:** Map from $\overline{A_{\mathsf{TM}}}$. If invalid input, output a machine that accepts everything (Σ^*, decidable). O/W, output M' that runs M on w. If M rejects, loop forever (Halting set = $\emptyset$). If M accepts, run the A_{TM} recognizer on input (Halting set = A_{TM}). **Key Insight:** To prove not in RE, map from $\overline{A_{\mathsf{TM}}}$. Tie the Accept/Reject behavior of M to Decidable ($\emptyset$, Σ^*) or Undecidable (A_{TM}) sets.
- **5. The Palindrome Property**: **Q:** Prove T (machines that accept w^R whenever they accept w) is not RE. **A:** Map from $\overline{A_{\mathsf{TM}}}$. Output M' that runs M on w. If it loops, $L(M') = \emptyset$ (satisfies T). If it accepts, M' acts as strict regular expression for $(01)^+$ (violates T because **10** is not accepted). **Key Insight:** Choose anchor languages wisely. Pick the simplest possible language that definitively proves or breaks the property you are testing.

III. Time Complexity & P vs. NP

- **6.** 🚩 **The Verifier Masterclass**: **Q:** Prove $PRIME\text{-}FACTOR \in \mathrm{NP} \cap \mathrm{coNP}$. **A:** The certificate is the prime factorization. NP verifier checks primality, verifies the product equals n, and any factor is $\leq m$. coNP verifier does the same math validation, but checks if *all* factors are $> m$. **Key Insight:** For coNP verifiers, outline that you cannot just "flip the output" of the NP verifier. You must fully validate that the certificate is mathematically honest before testing the NO condition.
- **7. Math Triviality vs. NP-Complete**: **Q:** $HALF\text{-}3SAT$ (satisfy half of clauses) vs $FIRST\text{-}HALF\text{-}3SAT$ (satisfy first half exactly). **A:** $FIRST\text{-}HALF$ is NPC via padding (add dummy clauses with new vars to second half). $HALF\text{-}3SAT$ is in P because a random guess satisfies **7/8** of 3-CNF clauses, so every valid formula always has a "YES" answer. The algorithm just checks syntax. **Key Insight:** Sometimes problems are in P not from clever search, but because math guarantees the answer is always YES.
- **8. Graph Padding Reduction**: **Q:** Reduce $CLIQUE$ to $HALF\text{-}CLIQUE$. **A:** If $k > n/2$, pad with isolated nodes. If $k < n/2$, pad with a fully connected clique of $n - 2k$ nodes, connected to every node in the original. **Key Insight:** Physically alter structure (padding with disconnected or fully connected nodes) to force the new parameter to equal the old one.

IV. Automata Mechanics & Closures

- **9. The Counter DFA**: **Q:** Build a counter DFA for $a^i b^j a^k c$ where $i + k > j$. **A:** Inverse counting: increment for b's and decrement for a's. Accept only if counter is negative (state **n**) when final **c** is read. **Key Insight:** When constrained by limited hardware, re-arrange the algebra of the inequality to fit the hardware.
- **10. Formal DFA Concatenation**: **Q:** Formally define DFA for $L(A) \cdot \{cc\} \cdot L(B)$. **A:** Define a new 5-tuple. Add bridge states to track the two **c**'s. Explicitly define q_{sink} to catch any premature **c**'s or invalid symbols for totally defined transitions. **Key Insight:** Formal definitions require absolute precision. If you expand the alphabet, you must define exactly where the new letters go for every single existing state.
- **11.** 🚩 **The Inequality Trap Pumping Lemma**: **Q:** Prove $a^j b^m$ where $j \neq 2m$ is not regular. **A:** Choose $s = a^{p!} b^{p!}$. Because $p!$ is a universal numerator, no matter y's length k, you can perfectly divide it to pump a's to $2(p!)$. **Key Insight:** When fighting inequality ($\neq$), pump *into* the equality. Use Factorials ($p!$) to guarantee perfect integer division when calculating the multiplier i. When the demon picks $|y| = k$, pump exactly $i = (p!/k) + 1$ times
- **12. State-Space Closure**: **Q:** Prove $\tilde{L} = \{uv \mid u^R v \in L\}$ is regular. **A:** Use original DFA states as halfway meeting points (p). Prove U_p regular (reversal closure), V_p regular, and concatenation L_p regular. Final language is finite union of all L_p. **Key Insight:** For infinite paths or word splits in a DFA, anchor proofs to the **finite states**. Finite unions of regular sub-languages preserve regularity.

V. Context-Free Languages & PDAs

- **13. CFG Reverse Engineering**: **Q:** Analyze $B \to aBb \mid C$ and $C \to aC \mid \varepsilon$, prove ambiguity. **A:** Language generated is $a^m b^n$ where $m \geq n$, because C rule injects a's into the *middle*, merging with left-hand a's. Ambiguity proven by drawing two distinct leftmost derivations for **aa**. **Key Insight:** CFG rules build from "outside in." Trace where variables physically drop terminals relative to boundaries.
- **14. CFL Pumping Lemma**: **Q:** Prove $a^i b^{2i-j} c^j$ ($2 \leq j < i$) is not CFL. **A:** Choose hair-trigger string $s = a^{p+3} b^{p+4} c^{p+2}$. Define cases strictly by where the window vxy falls (entirely in one section, or crossing a boundary). **Key Insight:** Define CFL pumping cases by the span of the vxy window, not contents of v and y. Eliminates loophole of v or y being empty. Ensure lengths are 1 unit apart so a single pump breaks inequality.

VI. The Foundations

- **15. Proof by Diagonalization**: **Q:** Prove a language like A_{TM} or $HALT_{\mathsf{TM}}$ is undecidable from scratch. **A:** Assume perfect decider H exists. Build spiteful dummy D taking $\langle M\rangle$. It feeds $\langle M, \langle M\rangle\rangle$ to H. D then conditionally does the *opposite* of H's prediction. Feed D its own source ($\langle D\rangle$); D accepts iff D rejects, a paradox. **Key Insight:** Diagonalization is weaponized spite. You assume a perfect observer exists, ask for its prediction, and intentionally defy it. Creates a paradox that proves observer cannot exist.

VII. Execution Mechanics & Exam Hacks

16. The "Dovetailing" (Parallel Simulation) Cheat Code

- **Problem & Solution:** You need to prove a language is Recognizable (in RE), but testing a machine might result in an infinite loop. 🚩 You cannot run M_1 to completion, then M_2. You must "Dovetail."** Run M_1 for 1 step. Run M_1, M_2 for 2 steps. Run M_1, M_2, M_3 for 3 steps.
- **When to use:** Building Enumerators from Recognizers, simulating multiple TMs, or proving union/intersection closures for RE.

17. Rice's Theorem: Prove L is Undecidable instantly

- **The Rule:** Any non-trivial property of the *language* recognized by a TM is undecidable. If P is non-trivial, then $LP = \{\langle M\rangle | L(M) \in P\}$ is Undecidable. 🚩 **The Trap:** The property MUST be about $L(M)$, not the physical machine M. $L(M1) = L(M2) \Leftrightarrow (M1 \in L \wedge M2 \in L) \vee (M1 \notin L \wedge M2 \notin L)$. -✅ *Undecidable (Rice applies):* "Does M accept '101'?" / "Is $L(M)$ regular?". 🚩 *Decidable (Rice FAILS):* "Does M have 5 states?" / "Does M halt in < 100 steps?"(checkable in finite time). **Extended Rice (Not in RE):** If the property P is true for the empty set ($L(M) = \emptyset$), then any language that *requires* accepting strings to satisfy P is **Not in RE** (because you can reduce from $\overline{A_{TM}}$ where the looping case results in $\emptyset$).

18. PDA & Stack Architecture Patterns

- **Equality ($i = j$):** Push A's, pop for B's.
- **Multiples ($i = 2j$):** Push TWO A's for every input A. Pop ONE for every B. (Or vice versa).
- **Inequality ($i \neq j$):** Non-determinism is required! Branch the machine at the start: Branch 1 guesses $i > j$, Branch 2 guesses $i < j$.
- **Adjacent Monotonicity ($k_{n-1} < k_n$):** To compare the final two blocks, push C's for Block $n - 1$. Use an ϵ-transition (no input consumed) to move to the final block. Pop C's for Block n. If the stack hits \$ while input still remains, transition to the accept state to guarantee $k_n > k_{n-1}$.

19. NP-Completeness Reduction Arsenal (Which to choose?)

Don't guess which NP-Complete problem to reduce from. Match the structure:

- **Graph Structure / Connecting things:** Reduce from `CLIQUE`, `INDEPENDENT-SET`, or `VERTEX-COVER`.
- *Hack:* $VC = \text{Total Nodes} \setminus IS$. If you find an Independent Set of size k, the remaining nodes form a Vertex Cover of size $V - k$.
- *The Holy Trinity Hack:* `CLIQUE` in G of size k $\iff$ `INDEPENDENT-SET` in $\overline{G}$ (the complement graph) of size k $\iff$ `VERTEX-COVER` in $\overline{G}$ of size $|V| - k$. You can trivially swap between these three!
- **Routing / Traversing:** Reduce from `HAM-CYCLE` or `HAM-PATH`.
- *The Cycle-to-Path Hack:* To reduce Cycle to Path, pick an arbitrary node v. Split it into two nodes, v_{start} and v_{end}. Connect v_{start} to all of v's original outgoing edges, and v_{end} to all incoming edges. A

Hamiltonian Path from v_{start} to v_{end} in the new graph strictly corresponds to a Hamiltonian Cycle in the old graph.

- **Sets / Grouping / Numbers:** Reduce from `SUBSET-SUM`, `PARTITION`, or `SET-COVER`.
- *The Target Forcing Hack:* If converting `SUBSET-SUM` (target T) to `PARTITION` (equal halves), add one or two massive "dummy" items to the set. Size them perfectly so that halving the new grand total mathematically *forces* the original items to sum to exactly T on one side.
- **Logic / Rigid Constraints:** Reduce from `3SAT`. (Variables = choices, Clauses = rules).
- *The 3SAT Gadget Hack:* Variables always map to a pair of mutually exclusive nodes (e.g., connected by an edge so you can only pick one: x or $\neg x$). Clauses map to a triangle/group of nodes where the problem's rules force you to select at least one "True" node to cover the clause.
- **The Padding Trick:** If reducing $k \to f(n)$ (e.g., `CLIQUE` to `HALF-CLIQUE`), physically pad the graph with disconnected nodes (to dilute the ratio) or a fully-connected subgraph (to artificially inflate the clique size).

VIII. The "Red Flag" Trap Checklist

If you see these phrases on the exam, alarms should go off in your head:

- 🚩 **"TM that halts in at most $|<M>|$ steps"**: *Trap:* It looks like the Halting Problem, but it's bound by a finite number. You can just simulate it for exactly that many steps. It is **Decidable**.
- 🚩 **"For all x..." vs "There exists an x..." in reductions**: *Trap:* If a property requires checking *all* strings (e.g., $L(M) = \Sigma^*$), it is usually not in RE (requires infinite verification). If it requires finding *one* string ($L(M) \neq \emptyset$), it is usually in RE (can dovetail to find it).

IX. Additional High-Yield Exam Problems

20. Decidability of DFA Graph Properties (`MMN6` - Q1)

- **Q:** Is $LONGEST_{DFA} = \{\langle A, w\rangle \mid w \text{ is the longest word in } L(A)\}$ decidable?
- **A:** Yes. A DFA has a longest word if and only if there are NO cycles on any path from the start state to an accept state. The decider checks for cycles (using DFS/BFS). If a cycle exists, it rejects. If no cycle exists, the DFA is a finite Directed Acyclic Graph (DAG), and it simply finds the longest path to an accept state and checks if it equals w. **Key Takeaway:** When checking decidability for DFAs, treat the DFA as a directed graph. Checking for infinite languages = checking for cycles.

21. NP-Complete Edge Gadgets (`SampleExam` - Q4a)

- **Q:** Reduce `VERTEX-COVER` to `DOMINATING-SET`.
- **A:** For every edge $e = (u, v)$ in the original graph, add a new "gadget" node x_e connected *only* to u and v. To dominate this isolated x_e, the dominating set is physically forced to pick either u, v, or x_e. Picking u or v is always strictly better as it covers other edges too. **Key Takeaway:** If you need to force an NP-Complete choice (like "pick at least one endpoint of every edge"), create a dead-end "gadget" node attached to those endpoints.

22. The $O(n)$ Multi-Tape Algorithm (`MMN5` - Q5)

- **Q:** Build a 2-tape TM that decides $D = \{ww\}$ in strictly $O(n)$ time.
- **A:** Tape 1 has two pointers: a "fast" pointer moving 2 steps per cycle, and a "slow" pointer moving 1 step. When the fast pointer hits the end, the slow pointer is exactly in the middle. Copy the second half to Tape 2, rewind Tape 1 to the start, and compare Tape 1 to Tape 2 simultaneously. **Key Takeaway:** To find the middle of a string in linear time without sweeping back and forth (which costs $O(n^2)$), use the "fast/slow pointer" trick on a multi-tape machine.

23. The Un-Recognizability of Loops (`moedA1` - Q3b)

- **Q:** Is $Loop(v) \setminus \{v\}$ recognizable? (Where $Loop(v)$ is all TMs that infinite-loop on a specific input v).
- **A:** No. $Loop(v)$ is mathematically equivalent to $\overline{HALT_{TM}}$ (just hardcoded for input v), which is known to be Not in RE. Subtracting a single finite string $\{v\}$ from an infinite non-RE set cannot magically make it recognizable. **Key Takeaway:** You cannot build a recognizer for machines that *don't* halt. Removing a finite number of edge cases from an un-recognizable set leaves it un-recognizable.

X. Closure for R/RE and Key Theorems

Op	R (Decidable)	RE
Union / Concat / *	✓ Yes	✓ Yes
Intersect	✓ Yes	✓ Yes
Complement	**✓ Yes**	**X NO**

Note: If RE were closed under complement, then $RE = R$ (This is False).

1. $R = RE \cap \text{co-}RE$ (If L and $\bar{L}$ are recognizable $\to$ decidable).
2. $A_{TM} \in RE \setminus R$ (Halting Problem).
3. $\overline{A_{TM}} \in \text{co-}RE \setminus RE$ (Not even recognizable!).

XI. CFG: Prefix/Suffix Logic

Language: $L = \{cw_1 \dots cw_k \mid k \geq 3, \exists i : \#_a(w_i) = 2\#_b(w_{i+2}) + 1\}$

Rule:

- $S \to PcMQ$ (Structure: Prefix, Triple, Suffix)
- $P, Q \to cWP \mid \varepsilon$ (Generates generic blocks cw)
- $M \to BaBaMAbA \mid BaBcWcA$ (Recursion: $2a \in w_i \leftrightarrow 1b \in w_{i+2}$)
- $W \to aW \mid bW \mid \varepsilon$ (Base: Adds last $+1a, w_{i+1}$)
- $A \to aA \mid \varepsilon, B \to bB \mid \varepsilon$ (Free chars: $b \in w_i, a \in w_{i+2}$)

M-Rule Breakdown:

- **Rec ($BaBaMAbA$):** Pairs $2 \times a$ in w_i with $1 \times b$ in w_{i+2} for each step.
- **Base ($BaBcWcA$):** Stops recursion, adds the required $+1$ 'a' to w_i, and generates w_{i+1}.
- **Vars (A, B):** Allow "uncounted" chars ($b \in w_i, a \in w_{i+2}$) to fulfill $w \in \{a, b\}^*$.

XII. CFG: Distant Dependency

Language: $L = \{dw_1 \dots w_n d \mid n \geq 4, \exists k \in [2, n-2] : |w_k| = \#_c(w_{k+2})\}$

Rules:

- $S \to dPdMQd$ (Structure: Ensure d delimiters)
- $P \to WdP \mid W$ (Prefix: ≥ 1 block $w_1 \dots w_{k-1}$)
- $Q \to dWQ \mid \varepsilon$ (Suffix: blocks after w_{k+2})
- $M \to aMVcV \mid bMVcV \mid cMVcV \mid dWdV$ (Match $|w_k|$ to $\#_c(w_{k+2})$)
- $W \to aW \mid bW \mid cW \mid \varepsilon$ (Any string in $\{a, b, c\}^*$)
- $V \to aV \mid bV \mid \varepsilon$ (Free a, b in w_{k+2}; no c's)

Logic: P ensures $k \geq 2$ by generating at least w_1. M recursively pairs any char in w_k with a 'c' in w_{k+2}. V allows non-'c' chars in w_{k+2}. Base of M creates middle block w_{k+1}.

XIII. CFG: Examples

$L_1 = \{w \in \{a, b\}^* \mid \#_b(w) \bmod 3 = 1\}$
$S \to aX \mid bS, \quad X \to aY \mid bX \mid \varepsilon, \quad Y \to aS \mid bY$
$L_2 = \{a^n b^n \mid n \geq 0\}$
$S \to aSb \mid \varepsilon$
$L_4 = \{a^{3n} b^{n+1} \mid n \geq 0\}$
$S \to aaaSb \mid b$
$L_5 = \{a^m b^k \mid m > k\}$
$S \to aSb \mid aS \mid a$
$L_6 = \{a^m b^k \mid m \neq k\}$
$S \to aSb \mid A \mid B, \quad A \to aA \mid a, \quad B \to bB \mid b$
$L_7 = \{a^m b^k c^{2m+k} \mid m, k > 0\}$
$S \to aScc \mid aXcc, \quad X \to bXc \mid bc$
$L_8 = \{a^m b^{m-k} c^k \mid m > k\}$
$S \to aSc \mid X, \quad X \to aXb \mid ab$
$L_9 = \{a^m b^k \mid k \leq m \leq 3k\}$
$S \to aSb \mid aaSb \mid aaaSb \mid \varepsilon$
$L = \{a^m b^k \mid m, k \geq 0, m \leq k \leq 3m\}$
$S \to aSb \mid aSbb \mid aSbbb \mid \varepsilon$
$L_{10} = \{a^m b^k \mid k < m < 2k\}$
$S \to aSb \mid aaSb \mid aaabb$
$L_{11} = \{a^m b^m a^k b^k \mid m, k \geq 0\}$
$S \to XX, \quad X \to aXb \mid \varepsilon$
$L_{12} = \{a^m b^s c^k d^t \mid m + s = k + t\}$
$S \to aSd \mid X \mid Y, \quad X \to aXc \mid Z, \quad Y \to bYd \mid Z, \quad Z \to bZc \mid \varepsilon$
$L_{13} = \{a^m b^s c^k d^t \mid m + t = s + k\}$
$S \to XY \mid MN, \quad M \to aXb \mid \varepsilon, \quad N \to bNd \mid Y, \quad Y \to cYd \mid \varepsilon,$
$X \to aXc \mid M$
$L_{14} = \{w \in \{a, b\}^* \mid w^R = w\}$
$S \to aSa \mid bSb \mid a \mid b \mid \varepsilon$
$L_{15} = \{w \in \{a, b\}^* \mid w^R \neq w\}$
$S \to aSa \mid bSb \mid aXb \mid bXa, \quad X \to aX \mid bX \mid \varepsilon$
$L_{16} = \{w \in \{a, b\}^* \mid w^R = w, \#_a(w) \bmod 3 = 1\}$
$S \to bSb \mid aXa \mid a, \quad X \to bXb \mid aYa, \quad Y \to bYb \mid aSa \mid b \mid \varepsilon$
$L_{17} = \{w \in \{a, b\}^* \mid \#_a(w) = \#_b(w)\}$
$S \to aSb \mid bSa \mid SS \mid \varepsilon$
$L_{18} = \{w \in \{a, b\}^* \mid \#_a(w) = \#_b(w) + 1\}$
$S \to XaX, \quad X \to aXbX \mid bXaX \mid \varepsilon$
$L_{19} = \{w \in \{a, b\}^* \mid \#_a(w) = 2 \cdot \#_b(w)\}$
$S \to aSaSbS \mid aSbSaS \mid bSaSaS \mid \varepsilon$
$L = \overline{\{ww \mid w \in \{a, b\}^*\}}$
$S \to S_1 \mid S_2, \quad S_1 \to TTS_1 \mid T, \quad S_2 \to XY \mid YX, \quad X \to TXT \mid a,$
$Y \to TYT \mid b, \quad T \to a \mid b$
$L = \overline{\{a^n b^n c^n \mid n \geq 0\}}$
$S \to S_1 \mid S_2 \mid S_3 \mid S_4 \mid S_5, \quad S_1 \to XbXaX, \quad X \to aX \mid bX \mid cX \mid \varepsilon,$
$S_4 \to MC, \quad M \to aMb \mid aA \mid bB, \quad A \to aA \mid \varepsilon, \quad B \to bB \mid \varepsilon,$
$C \to cC \mid \varepsilon$
$L_{20} = \{a^m b^n c^p \mid m + 2n \geq p\}$
$S \to aSc \mid aS \mid A, \quad A \to bAcc \mid bAc \mid bA \mid \varepsilon$
$L_{21} = \{a^m b^n \mid 3m \leq 5n \leq 4m\}$
$S \to a^4 Ab^3 \mid a^3 Ab^2 \mid a^7 Ab^5 \mid a^6 Ab^4 \mid A, \quad A \to a^5 Ab^3 \mid a^5 Ab^4 \mid \varepsilon$